gpeyt1-21 · Borrow quantity

GEO PE · Long Calculation Problems I — Topic 5: Earth Structures, Ground Improvement & Pavements

Problem

A highway embankment needs 85,000 yd³ of compacted fill at 95% of standard Proctor γd,max = 118.5 pcf. The borrow pit soil has an in-place moist unit weight of 112 pcf at w = 14%. The loose (truck) unit weight is 96 pcf and each truck carries 14 loose yd³. Ignore haul losses.

Most nearly, the volume of borrow (bank measure) required is:

A. 85,436 yd³
B. 102,524 yd³
C. 74,181 yd³
D. 97,397 yd³

Given equations / data

> Dry solids are conserved: Vborrow = Vfill × γd,fill / γd,borrow; γd = γ/(1 + w)

Worked solution

1. γd,fill = 0.95 × 118.5 = 112.6 pcf
2. γd,borrow = 112/1.14 = 98.2 pcf
3. Vborrow = 85,000 × 112.6/98.2 = 97,397 yd³

Final answer

97,397 yd³

Why the other values are wrong

- **85,436 yd³** — used moist borrow unit weight as dry
- **102,524 yd³** — forgot the 95% factor
- **74,181 yd³** — inverted the density ratio

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.3 Weight-Volume Relationships, p. 133; §2.3.1 earthwork unit weights (γB, γC)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=v9reh5FJnUc (no creator values copied).

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